

Null Responses and Positive Energy in Motion–Staticity Geometry: Hermitian Cones, Sen–Witten Lifts, and Quantum Constraints

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Abstract

The motion–staticity (MS) compactification turns a future-timelike translation covector into a point of a closed three-ball whose boundary is the sphere of projective null directions. This paper studies the positive-energy problem on that geometry. The normalized MS ball is first written as the trace-one section of the positive cone of 2×2 Hermitian matrices, with staticity $S = 2\sqrt{\det \rho}$. Rank-one boundary projectors are the extreme points of the cone, and the MS observer response is the trace pairing between a translation matrix and such normalized states. In particular, positivity of all null responses $F(n) = E - p \cdot n$ is equivalent to positivity of the Hermitian translation matrix and hence to $E \geq |p|$.

For Einstein initial data, a direct constraint identity proves this directional positivity on maximal conformally flat slices. A background-free orthonormal-frame current then removes the local curvature and nonmaximality obstructions under a special frame/lapse gauge, and a div–curl implicit-function argument constructs that gauge for sufficiently small asymptotically flat perturbations of Euclidean data. The gauge is exactly the real form of the Sen–Witten equation. Within a fixed one-parameter frame ansatz, universal local positivity selects the Sen–Witten trace coefficient, while an explicit local-phase counterexample shows that phase-blind null-response data cannot select the full spinor equation. Thus the projective null direction must be distinguished from its spinorial lift.

The timelike translation matrix also induces a Hermitian metric on the basic boundary spinor line. Its Chern curvature is $-\frac{1}{2}m^2 F(n)^{-2}d\Omega$, and the helicity- h power has curvature $-hm^2 F(n)^{-2}d\Omega$ with unchanged Chern number $-2h$. This identifies the null response as the conformal weight relating the observer-dependent visual sphere to the standard projective boundary.

The final part records quantum consistency tests rather than a quantum ADM theorem. On a specified positive massive shell one obtains an exact operator identity $\hat{F}_n \geq (c/2)\hat{q}$, together with a collective-staticity variance formula and a controlled attractive two-body example. By contrast, full additive Weyl-shift covariance is incompatible with a semibounded physical energy, classical positive squares do not survive arbitrary quantization orderings, and pointwise classical energy conditions do not become universal quantum local-positivity conditions. The results isolate the structures that a future constrained quantum boundary theory must supply.

1 Introduction

The MS construction separates two questions that are often merged in relativistic kinematics. The first is how a local operational response determines a soldered Lorentzian geometry. The second is how the resulting geometry supports total translation charges for an asymptotically

isolated gravitating system. In the classical boundary-completion analysis, the latter question leads to a covector P_∞^{MS} on the asymptotic translation space and, after the empirical Einstein normalization is fixed, to the standard ADM four-momentum. That construction does not by itself prove that the total covector lies in the future cone.

The positive-energy problem is therefore naturally phrased as a boundary response problem. If $P = (E, p)$ is a translation covector, an asymptotic future null direction $n \in S^2$ probes the scalar

$$F(n) = E - p \cdot n. \quad (1)$$

The familiar inequality $E \geq |p|$ is equivalent to $F(n) \geq 0$ for every unit n . In MS variables this boundary test is the extremal limit of the massive observer response. The main purpose of the present paper is to make the geometry of that statement explicit and then to examine what, classically and quantum mechanically, can enforce it.

Three ingredients are kept distinct throughout. First, the Lorentz cone and its spinorial realization are kinematical. Second, positivity for gravitational initial data uses the Einstein constraints together with an energy condition. Third, the Sen–Witten equation is a spinorial selection equation; it is not supplied by the primitive scalar response alone. This separation is important because the same formal objects can otherwise be over-interpreted. The positive-mass theorem and the Witten/Nester mechanisms are established results [**SchoenYau1979**, **Witten1981**, **NesterTung1994**]. What is developed here is their organization in MS response variables, several explicit frame identities and restricted selection tests, and the resulting constraints on a prospective quantum boundary theory.

The paper is arranged as follows. [Section 2](#) gives the Hermitian-cone formulation of the closed MS ball and the convex decomposition of massive states into null extreme points. [Section 3](#) derives positive classical response identities, first in a conformally flat maximal class and then with a background-free orthonormal frame. The latter identity yields a small-data positive-energy theorem without conformal flatness or maximality. [Section 4](#) identifies the frame gauge with the Sen–Witten equation, derives a restricted coefficient-selection result, and proves a phase-blind obstruction. [Section 5](#) relates the timelike response function to the Hermitian and Chern geometry of the boundary spinor line. [Section 6](#) collects exact quantum-shell identities and no-go statements that delimit a possible quantum ADM completion. We use signature $(-, +, +, +)$ in the gravitational sections and keep $c = 1$ until the quantum discussion.

2 The normalized Lorentz–Hermitian cone

2.1 The MS ball as a trace-one positive cone

Let σ_i be the Pauli matrices. To a dimensionless motion vector $D \in \mathbb{R}^3$ associate

$$\rho(D) = \frac{1}{2}(\mathbf{1} + D \cdot \sigma). \quad (2)$$

Its eigenvalues are $(1 \pm |D|)/2$. Hence $\rho(D)$ is positive definite for $|D| < 1$, positive semidefinite for $|D| \leq 1$, and has unit trace. If the MS staticity variable is defined by $|D|^2 + S^2 = 1$ with $S \geq 0$, then

$$\det \rho(D) = \frac{1}{4}(1 - |D|^2) = \frac{S^2}{4}, \quad S = 2\sqrt{\det \rho(D)}. \quad (3)$$

Thus the closed MS state space is canonically identified with

$$\bar{\mathcal{B}}_{\text{MS}} = \{\rho = \rho^\dagger \geq 0 : \text{Tr } \rho = 1\},$$

the trace-one section of the positive Hermitian cone. The interior is the rank-two locus, while the projective null boundary is the rank-one locus.

Every rank-one point can be written as

$$\Pi_n = \frac{1}{2}(\mathbf{1} + n \cdot \sigma) = \frac{\zeta \zeta^\dagger}{\zeta^\dagger \zeta}, \quad |n| = 1, \quad (4)$$

where $\zeta \in \mathbb{C}^2 \setminus \{0\}$ is defined only up to multiplication by a nonzero complex number. Consequently $\partial \bar{\mathcal{B}}_{\text{MS}} \simeq \mathbb{CP}^1 \simeq S^2$. This is the projective-null boundary used in the companion spin analysis. It is a space of null rays, not a fixed-energy massless momentum orbit.

Proposition 2.1 (Convex decomposition along a motion ray). *For $D = dn$ with $0 \leq d \leq 1$ and $|n| = 1$,*

$$\rho(dn) = \frac{1+d}{2}\Pi_n + \frac{1-d}{2}\Pi_{-n}. \quad (5)$$

In particular, the closed MS ball is the convex hull of its rank-one projective-null boundary.

Proof. Insert $\Pi_{\pm n} = \frac{1}{2}(\mathbf{1} \pm n \cdot \sigma)$ into the right-hand side of (5). The identity follows immediately from (2). $\square$

The decomposition is elementary, but it will later explain why positivity on the null boundary controls every massive observer response. In this sense the projective null states are the extremal probes of the normalized cone.

2.2 Translation covectors as Hermitian forms

Let $P = (E, p)$ be a translation covector and define

$$\mathbf{P} = E\mathbf{1} - p \cdot \sigma. \quad (6)$$

The sign convention is chosen so that the rank-one projector (4) gives

$$\text{Tr}(\mathbf{P}\Pi_n) = E - p \cdot n = F(n). \quad (7)$$

For a massive MS observer state $\rho(D)$, the same pairing gives

$$\text{Tr}(\mathbf{P}\rho(D)) = E - p \cdot D = S \mathcal{E}(D, S), \quad (8)$$

where $\mathcal{E}(D, S) = (E - p \cdot D)/S$ is the usual observer response for $S > 0$.

Proposition 2.2 (Null tests and Hermitian positivity). *The following statements are equivalent:*

- (i) $F(n) \geq 0$ for every unit vector n ;
- (ii) $\mathbf{P} \geq 0$ as a Hermitian matrix;
- (iii) $E \geq |p|$.

If these conditions hold, then $\text{Tr}(\mathbf{P}\rho) \geq 0$ for every $\rho \in \bar{\mathcal{B}}_{\text{MS}}$.

Proof. The eigenvalues of $\mathbf{P}$ are $E \pm |p|$. Thus (ii) and (iii) are equivalent. Rank-one projectors exhaust the extreme rays of the dual test set, so (i) is equivalent to nonnegativity of the quadratic form $\zeta^\dagger \mathbf{P} \zeta$ on every spinor line, hence to (ii). The final assertion follows from positivity of the trace pairing of two positive Hermitian matrices, or directly from Proposition 2.1. $\square$

For $D = dn$, Proposition 2.1 and (8) yield

$$S \mathcal{E}(D, S) = \frac{1+d}{2}F(n) + \frac{1-d}{2}F(-n). \quad (9)$$

This is the convex-geometric origin of the “null estimates imply massive estimates” relation used below.

2.3 Timelike points and the hyperbolic interior

If $\mathbf{P}$ is positive definite, set $m = \sqrt{\det \mathbf{P}} = \sqrt{E^2 - |p|^2}$ and $U = P/m$. Positive determinant fixes the timelike interior, while $m \rightarrow 0$ approaches the rank-one null boundary. Up to an overall scale, the interior may be realized as the symmetric space $SL(2, \mathbb{C})/SU(2)$ via positive Hermitian matrices of determinant one; see, for example, the two-spinor description of Lorentz geometry in [PenroseRindler1984]. The normalized trace-one section used here is the compact ball model of the same timelike data.

The boundary function $F(n)$ also determines the observer-dependent conformal representative of the celestial sphere. This will be used in Section 5; at this stage the only point needed is that $F(n)$ is the dual pairing between a timelike interior point and a null extreme point.

3 Classical positive-response identities

3.1 Conventions and asymptotic charges

Let (Σ, h, k) be an asymptotically flat initial data set with future unit normal N . We use the sign convention $k = -K$ relative to the extrinsic curvature convention of the companion MS-ADM paper. Write $\tau = \text{tr}_h k$ and define matter fields $\epsilon = T(N, N)$ and $j_i = -T(N, \partial_i)$. The Einstein constraints are

$$R_h + \tau^2 - |k|_h^2 = 2\kappa\epsilon, \quad D_j(k^{ij} - \tau h^{ij}) = \kappa j^i, \quad (10)$$

with $\kappa = 8\pi G$ in units $c = 1$. The dominant energy condition (DEC) is $\epsilon \geq |j|_h$.

For asymptotically Cartesian coordinates the ADM energy and physical spatial momentum are

$$E = \frac{1}{2\kappa} \lim_{r \rightarrow \infty} \int_{S_r} (\partial_j h_{ij} - \partial_i h_{jj}) \nu^i dS, \quad p_i = \frac{1}{\kappa} \lim_{r \rightarrow \infty} \int_{S_r} (k_{ij} - \tau h_{ij}) \nu^j dS. \quad (11)$$

The directional quantity to be controlled is again $F(n) = E - p \cdot n$.

3.2 A conformally flat maximal identity

Assume first that $\Sigma \simeq \mathbb{R}^3$ and

$$h_{ij} = \lambda^2 \delta_{ij} = \psi^4 \delta_{ij}, \quad \lambda = \psi^2 > 0, \quad \tau = 0. \quad (12)$$

Define the Euclidean trace-free tensor $A^{ij} = \psi^{10} k^{ij} = \lambda^5 k^{ij}$. The constraints reduce to

$$\Delta \lambda = -\frac{1}{4} \lambda^{-3} |A|^2 - \frac{\kappa}{2} \epsilon \lambda^3 + \frac{1}{2\lambda} |\nabla \lambda|^2, \quad \partial_j A^{ij} = \kappa \lambda^5 j^i. \quad (13)$$

Assume $\lambda = 1 + \sigma/r + O_2(r^{-2})$, $A = O_1(r^{-2})$, and sufficient integrability of the matter sources. Then

$$E = -\frac{2}{\kappa} \lim_{r \rightarrow \infty} \int_{S_r} \partial_\nu \lambda dS = \frac{8\pi\sigma}{\kappa}, \quad p_i = \frac{1}{\kappa} \lim_{r \rightarrow \infty} \int_{S_r} A_{ij} \nu^j dS. \quad (14)$$

For a Euclidean unit vector n and a vector v , put

$$\mathbf{S}(n, v)_{ij} = \frac{1}{2} (n_i v_j + n_j v_i) - \frac{1}{3} (n \cdot v) \delta_{ij}$$

and define the vector field $\mathcal{F}_n = -2\lambda^{-2} \nabla \lambda - \lambda^{-3} A n$. The algebraic identities $A : \mathbf{S}(n, v) = A(n, v)$ and $|\mathbf{S}(n, v)|^2 = \frac{1}{2} |v|^2 + \frac{1}{6} (n \cdot v)^2$ give the following divergence formula.

Lemma 3.1 (Direct constraint identity). *Under (13),*

$$\begin{aligned} \text{div } \mathcal{F}_n &= \frac{1}{2} \lambda^{-5} |A + 3\lambda \mathbf{S}(n, \nabla \lambda)|^2 + \frac{3}{4} \lambda^{-3} (|\nabla \lambda|^2 - (n \cdot \nabla \lambda)^2) \\ &\quad + \kappa \lambda (\epsilon - \lambda n^i j_i). \end{aligned} \quad (15)$$

Every term on the right is nonnegative under DEC.

Proof. Differentiating the first part of $\mathcal{F}_n$ and using the Hamiltonian constraint gives

$$\operatorname{div}(-2\lambda^{-2}\nabla\lambda) = \frac{1}{2}\lambda^{-5}|A|^2 + 3\lambda^{-3}|\nabla\lambda|^2 + \kappa\lambda\epsilon.$$

The momentum constraint gives $\operatorname{div}(-\lambda^{-3}An) = 3\lambda^{-4}A(n, \nabla\lambda) - \kappa\lambda^2 n^i j_i$. Completing the square with the stated identities yields (15). Finally $|j|_h = \lambda|j|_\delta$, so $\epsilon - \lambda n^i j_i \geq \epsilon - |j|_h \geq 0$. $\square$

Theorem 3.2 (Restricted spacetime positive energy). *Let smooth data on $\mathbb{R}^3$ satisfy (12), the asymptotic conditions above, the Einstein constraints, and DEC. Then $E \geq |p|$. If equality holds, then $\lambda = 1$, $k = 0$, $\epsilon = j = 0$, and $E = |p| = 0$.*

Proof. There is no inner boundary, so the divergence theorem and (14) give

$$\begin{aligned} E - p \cdot n &= \frac{1}{2\kappa} \int_{\mathbb{R}^3} \lambda^{-5} |A + 3\lambda S(n, \nabla\lambda)|^2 d^3x \\ &\quad + \frac{3}{4\kappa} \int_{\mathbb{R}^3} \lambda^{-3} |\nabla\lambda - (n \cdot \nabla\lambda)n|^2 d^3x + \int_{\mathbb{R}^3} \lambda(\epsilon - \lambda n^i j_i) d^3x. \end{aligned} \quad (16)$$

Taking $n = p/|p|$ when $p \neq 0$ proves $E \geq |p|$. If equality holds, each nonnegative term vanishes. The transverse gradient vanishes, and asymptotic flatness forces $\lambda \equiv 1$ along lines perpendicular to n . The first square then gives $A = 0$ and the scalar constraint gives $\epsilon = 0$; DEC implies $j = 0$. $\square$

The same identity applies to standard puncture ends if the inner fluxes vanish under the usual asymptotics. Bowen–York and Brill–Lindquist data provide familiar examples of this setting [BowenYork1980, BrandtBruegmann1997, FrauendienerNesterSzabados2011].

The weights have a direct MS interpretation. The lapse and shift $L = \lambda^{-2}$ and $Y^i = \lambda^{-3}n^i$ define the future-null field $\ell_n = LN + Y$. Since $|Y|_h = L$, its soldered direction is proportional to $(1, n)$, while the matter contribution in (16) is exactly $\int T(N, \ell_n) dV_h$. The projective null direction is therefore the extremal observer probe selected by the cone geometry of Section 2.

3.3 Why the conformal proof does not immediately generalize

Two obstructions appear if one simply deforms the preceding calculation. If $\tau \neq 0$, the corrected conformal flux acquires an additional term $-(1/3)\lambda\tau^2$ with no sign supplied by DEC. If $h = \psi^4\bar{h}$ with nonflat $\bar{h}$, the corresponding identity contains reference-curvature and derivative terms whose signs are likewise uncontrolled. These failures show that the soldering variables do not remove the intrinsic curvature problem by themselves.

A scalar route can instead use spacetime-harmonic functions. The established analysis of Hirsch, Kazaras and Khuri, building on the Riemannian work of Bray, Kazaras, Khuri and Stern, gives a global positive-mass argument after controlling level-set curvature and topology [HirschKazarasKhuri2022, BrayKazarasKhuriStern2022]. The purpose here is different: we seek a frame identity that retains the null-response interpretation while exposing its relation to the Witten/Nester mechanism.

3.4 A background-free orthonormal-frame current

Let (e_1, e_2, e_3) be an oriented h -orthonormal frame and write

$$\Gamma_{abc} = h(D_{e_a}e_b, e_c) = \epsilon_{bcd}M_{ad}, \quad d_a = \operatorname{div}_h e_a = \epsilon_{abc}M_{bc}. \quad (17)$$

Set $t = \operatorname{tr} M$, $H = \operatorname{TF} \operatorname{sym} M$, and $C = -2 \sum_a d_a e_a$. A standard frame calculation gives

$$R_h = \operatorname{div}_h C + \frac{1}{2}|d|^2 - |H|^2 + \frac{2}{3}t^2. \quad (18)$$

No reference metric enters this identity.

Choose $a = e_3$ and a positive scalar L , and put $u = \nabla_h \log L$. Impose the four real gauge conditions

$$d_a = -e_a(\log L) + \tau \delta_{a3}, \quad t = 0. \quad (19)$$

These are conditions to be solved, not properties of an arbitrary soldering frame. Define

$$\begin{aligned} \mathbf{S}(a, u)_{ij} &= \frac{1}{2}(a_i u_j + a_j u_i) - \frac{1}{3}u(a)\delta_{ij}, \\ \mathbf{U}_a(H)_{ij} &= \frac{1}{2}(\epsilon_{3jd}H_{id} + \epsilon_{3id}H_{jd}), \end{aligned} \quad (20)$$

and the current

$$J = \frac{L}{2}C - L(k - \tau h)(a, \cdot)^\sharp. \quad (21)$$

Proposition 3.3 (Conditional covariant positive current). *If the constraints and (19) hold, then*

$$\begin{aligned} \operatorname{div}_h J &= L \left\{ \frac{1}{2} \left| k_0 - \mathbf{U}_a(H) - \frac{3}{2}\mathbf{S}(a, u) + \frac{\tau}{2}\mathbf{S}(a, a) \right|^2 \right. \\ &\quad \left. + \frac{3}{4} \left[H(a, a)^2 + \left| H(a, \cdot)^\sharp - H(a, a)a - \frac{1}{2}a \times u \right|^2 \right] + \kappa(\epsilon - j(a)) \right\}, \end{aligned} \quad (22)$$

where $k_0 = k - \frac{1}{3}\tau h$. The right-hand side is nonnegative under DEC for arbitrary τ .

Proof. With $t = 0$, decompose $M_{ij} = H_{ij} + \frac{1}{2}\epsilon_{ijc}d_c$. Differentiating (21), using (18), the Einstein constraints, and $d = -u + \tau a$, gives

$$\begin{aligned} \frac{\operatorname{div} J}{L} &= \frac{1}{2}|k_0|^2 - k_0 : \left(\mathbf{U}_a(H) + \frac{3}{2}\mathbf{S}(a, u) - \frac{\tau}{2}\mathbf{S}(a, a) \right) + \frac{1}{2}|H|^2 + \frac{3}{4}|u|^2 + \frac{\tau^2}{12} - \frac{\tau}{2}u(a) \\ &\quad + \kappa(\epsilon - j(a)). \end{aligned}$$

The identities $\mathbf{U}_a(H) : \mathbf{S}(a, a) = 0$, $\mathbf{S}(a, u) : \mathbf{S}(a, a) = 2u(a)/3$, and $|\mathbf{S}(a, a)|^2 = 2/3$ place the entire trace dependence inside the first square. The remaining quadratic form satisfies

$$\frac{1}{2}|H|^2 + \frac{3}{4}|u|^2 - \frac{1}{2} \left| \mathbf{U}_a(H) + \frac{3}{2}\mathbf{S}(a, u) \right|^2 = \frac{3}{4} \left[H(a, a)^2 + \left| H(a)_\perp - \frac{1}{2}a \times u \right|^2 \right],$$

which gives (22). $\square$

In the gauge (19), J simplifies to $J = \nabla_h L - Lk(a, \cdot)^\sharp$. The associated future-null probe is $\ell = L(N + a)$, and its matter response is $T(N, \ell) = L(\epsilon - j(a))$.

For an asymptotically Cartesian frame with decay exponent $q > 1/2$, the antisymmetric part of the leading frame perturbation contributes zero closed-sphere flux. If $L = 1 + O_1(r^{-q})$ and $a \rightarrow n$, then

$$\frac{1}{\kappa} \lim_{r \rightarrow \infty} \int_{S_r} h(J, \nu_h) dS_h = E - p \cdot n. \quad (23)$$

Thus a global regular solution of (19) converts every null response into a nonnegative bulk integral.

3.5 A small-data global construction

The global solvability of (19) is nontrivial for general data. A perturbative construction is nevertheless available near Euclidean space. Fix $q \in (1/2, 1)$ and weighted Hölder spaces $C_{-q}^{m,\alpha}$. Let $e_a^0 = (h^{-1/2})_a^i \partial_i$ be the positive symmetric square-root frame, rotate it by $e_a = (\exp \Omega(\theta))_a^b e_b^0$, and write $L = e^f$. The nonlinear gauge map is

$$G_{h,\tau}(\theta, f) = (d_a + e_a f - \tau \delta_{a3}, \text{tr } M).$$

At Euclidean data its linearization is the div-curl operator

$$\mathcal{L}_0(\theta, f) = (\text{curl } \theta + \nabla f, \text{div } \theta). \quad (24)$$

For prescribed (F, g) one solves $\Delta f = \text{div } F$ and $\Delta \theta = \nabla g - \text{curl } F$. The Euclidean weighted Laplacian is invertible at weights $-q \in (-1, 0)$, and the remaining decaying harmonic errors vanish. Bartnik's weighted asymptotic framework provides a standard analytic setting for these estimates [Bartnik1986].

Theorem 3.4 (Small-data energy-momentum inequality). *Let smooth complete Einstein initial data on $\mathbb{R}^3$ satisfy DEC, have well-defined ADM charges, and obey $h - \delta \in C_{-q}^{m+1,\alpha}$ and $\tau \in C_{-q-1}^{m,\alpha}$ with sufficiently small norms, where $q \in (1/2, 1)$. Assume the corresponding decay of k and integrability of the matter densities. Then $E \geq |p|$, without assuming conformal flatness or maximality.*

Proof. The bounded inverse of (24) and the Banach implicit-function theorem give a global small solution of (19), uniformly in the asymptotic direction n . Integrating (22) and using (23) gives $E - p \cdot n \geq 0$ for every n . Proposition 2.2 then gives $E \geq |p|$. $\square$

The theorem is deliberately local in field space. It does not replace the global positive-mass theorem or its rigidity statement, and it does not prove global gauge continuation at arbitrary field strength.

4 The Sen–Witten lift and what positivity selects

4.1 Exact relation between the frame gauge and the spinor equation

The frame identity above is not an independent replacement for the Witten mechanism. Let σ_a be Pauli matrices adapted to the orthonormal frame and let

$$\nabla_a \chi = e_a \chi - \frac{i}{2} M_{ad} \sigma_d \chi. \quad (25)$$

For an adapted spinor $\chi = \sqrt{L} \chi_0$ with constant components and $\sigma_3 \chi_0 = \chi_0$, one finds

$$\sigma^a \nabla_a \chi = \frac{1}{2} [(u + d)_a \sigma^a - i \text{tr } M] \chi. \quad (26)$$

The spacetime spin lift restricted to the hypersurface is the Sen connection

$$\nabla_a^{\text{Sen}} \chi = \nabla_a^h \chi - \frac{1}{2} k_{ab} \sigma^b \chi, \quad D_{\text{Sen}} \chi := \sigma^a \nabla_a^{\text{Sen}} \chi = \left(\sigma^a \nabla_a^h - \frac{\tau}{2} \right) \chi, \quad (27)$$

where symmetry of k and $\sigma^a \sigma^b = \delta^{ab} \mathbf{1} + i \epsilon^{abc} \sigma_c$ were used. Hence (19) is exactly the real frame form of $D_{\text{Sen}} \chi = 0$. The relation between Witten spinors and special orthonormal frames is established in the literature [Nester1992, NesterTung1994, FrauendienerNesterSzabados2011].

This identification clarifies the role of the projective boundary. A null direction fixes the rank-one projector Π_n , or equivalently a complex spinor line $[\zeta] \in \mathbb{CP}^1$. The Sen–Witten problem requires an actual section of the spinor bundle with a calibrated asymptotic value, not merely the projective line. The distinction will be essential below.

4.2 A restricted coefficient-selection test

The spinorial Gauss restriction fixes the coefficient $1/2$ in (27) once the spacetime connection and spin lift are specified. One may nevertheless ask whether positivity sees the same coefficient within the frame current itself. Replace (19) by

$$d = -u + \gamma\tau a, \quad t = 0, \quad (28)$$

while keeping the current (21) fixed. For an adapted spinor this corresponds to $D_b\chi = 0$ with $D_b = \sigma^a \nabla_a^h - b\tau$ and $\gamma = 2b$.

Proposition 4.1 (Restricted coefficient selection). *Under the constraints and (28),*

$$\begin{aligned} \frac{\operatorname{div}_h J}{L} = & \frac{1}{2} \left| k_0 - \mathcal{U}_a(H) - \frac{3}{2}\mathcal{S}(a, u) + \frac{\gamma\tau}{2}\mathcal{S}(a, a) \right|^2 \\ & + \frac{3}{4} \left[H(a, a)^2 + \left| H(a)_\perp - \frac{1}{2}a \times u \right|^2 \right] + \kappa(\epsilon - j(a)) - \frac{(\gamma - 1)^2}{3}\tau^2. \end{aligned} \quad (29)$$

Within this fixed current and one-parameter gauge family, algebraic nonnegativity for arbitrary local data satisfying DEC requires $\gamma = 1$, equivalently $b = 1/2$.

Proof. The calculation is the same as in Proposition 3.3 until the trace terms are completed. The first square contributes $\gamma^2\tau^2/12 - \gamma\tau u(a)/2$. The remaining trace coefficient is $-(\gamma - 1)^2/3$. If $\gamma \neq 1$, one may set $H = u = 0$, choose nonzero τ , and set $k_0 = -\gamma\tau\mathcal{S}(a, a)/2$ and $\epsilon = j = 0$ in the reduced algebraic expression. Both squares vanish while the final term is negative. $\square$

The proposition is not a uniqueness theorem for all covariant spinor gauges. It only states that the Sen–Witten coefficient is singled out by universal positivity within the specified current and ansatz. More general gauge choices and energy expressions have long been studied, notably by Parker [Parker1985].

4.3 Why phase-blind null data are insufficient

The previous result concerns one coefficient; it does not derive the full first-order equation from a scalar response principle. The obstruction is already visible on flat data.

Normalize χ_0 to unit norm and put $\chi = \sqrt{L}\chi_0$ in an adapted frame. If $r = d + u - \tau a$, then (26) gives

$$D_{\text{Sen}}\chi = \frac{\sqrt{L}}{2}(r_a\sigma^a - it)\chi_0, \quad |D_{\text{Sen}}\chi|^2 = \frac{L}{4}(|r|^2 + t^2). \quad (30)$$

The full equation imposes four real conditions. By contrast, the phase-invariant bilinears determine only $L = \chi^\dagger\chi$ and the null direction $La_i = \chi^\dagger\sigma_i\chi$. Compatibility gives the single projection

$$2\operatorname{Re}(\chi^\dagger D_{\text{Sen}}\chi) = \operatorname{div}_h(La) - \tau L. \quad (31)$$

Thus the spatial flux balance $\operatorname{div}_h(La) = \tau L$ captures only one real component of the spinor equation.

Proposition 4.2 (Phase-blind comparison obstruction). *A selection principle depending only on h , k , the phase-invariant bilinears L and a , their null current, and other local Hermitian bilinears cannot by itself select precisely the Sen–Witten spinors with a prescribed constant asymptotic value.*

Proof. On flat asymptotically Euclidean data take a constant unit spinor χ_0 and a smooth real compactly supported function f . The spinors χ_0 and $\chi_f = e^{if}\chi_0$ have identical L , a , null current, and asymptotic value. Nevertheless

$$D_{\text{Sen}}\chi_f = ie^{if}\sigma^a(\partial_a f)\chi_0, \quad |D_{\text{Sen}}\chi_f|^2 = |\nabla f|^2,$$

whereas $D_{\text{Sen}}\chi_0 = 0$. A phase-blind rule cannot distinguish the two inputs. $\square$

The result does not exclude a phase-sensitive comparison algebra, nor does it identify the momentum-space Berry phase of a helicity bundle with the spatial phase of a Sen–Witten field. It shows only that the projective null ray is not sufficient data for the positive-energy spinor lift. A full lift requires the complex line together with a phase-sensitive connection and a global elliptic selection problem.

A least-squares functional $\int_{\Sigma} |D_{\text{Sen}}\chi|^2 dV_h$ does not by itself solve this problem: compactly supported variation yields $D_{\text{Sen}}^\dagger D_{\text{Sen}}\chi = 0$, while one must still prove that the boundary-compatible minimum is attained at zero and that the resulting spinor is globally suitable. These analytic issues are part of the established Witten/Parker–Taubes theory [**ParkerTaubes1982**] and of later studies of spinorial energy functionals [**AmmannWeissWitt2016**].

5 Boundary spinor metrics and response-weighted Chern curvature

The companion spin paper analyzes helicity line bundles over the projective null boundary. The cone geometry gives an additional relation: a timelike translation covector determines a natural Hermitian metric on the basic boundary spinor line, and the null response is exactly its pointwise norm.

Let $\mathcal{H} \rightarrow \mathbb{CP}^1$ denote the tautological spinor line with the convention $c_1(\mathcal{H}) = -1$. If $[\zeta] \in \mathbb{CP}^1$ and $v \in \mathcal{H}_{[\zeta]}$, define

$$h_P(v, v) = v^\dagger P v. \quad (32)$$

For a future-timelike P , this is a positive Hermitian metric. If ζ is normalized and $n = \zeta^\dagger \sigma \zeta$, then $h_P(\zeta, \zeta) = F(n)$.

Proposition 5.1 (Response-weighted Chern curvature). *Let P be positive definite with $m^2 = \det P = E^2 - |p|^2$. The Chern curvature of $\mathcal{H}$ induced by (32), written as a real two-form in the present helicity convention, is*

$$f_{1/2, P} = -\frac{1}{2} \frac{m^2}{F(n)^2} d\Omega. \quad (33)$$

For the helicity- h line $L_h \simeq \mathcal{H}^{\otimes 2h}$,

$$f_{h, P} = -h \frac{m^2}{F(n)^2} d\Omega, \quad \frac{1}{2\pi} \int_{S^2} f_{h, P} = -2h. \quad (34)$$

Proof. Use the affine coordinate z on $\mathbb{CP}^1$ with holomorphic frame $s(z) = (1, z)^T$. Put $H_P(z) = s(z)^\dagger P s(z)$. A direct 2×2 calculation gives

$$\partial_z \partial_{\bar{z}} \log H_P = \frac{\det P}{H_P^2}. \quad (35)$$

The normalized spinor is $\zeta = s/\sqrt{1+|z|^2}$, so $F(n) = H_P/(1+|z|^2)$ and $d\Omega = 2i dz \wedge d\bar{z}/(1+|z|^2)^2$. With the convention $f_{1/2, P} = -i\partial\bar{\partial} \log H_P$, (33) follows. Tensor power multiplies the curvature by $2h$, giving (34). The integral is invariant under the induced $SL(2, \mathbb{C})$ Möbius action and equals the rest-frame value $-4\pi h$. $\square$

The conformal factor in (34) also defines

$$g_P = \frac{m^2}{F(n)^2} d\Omega^2, \quad (36)$$

which is the round metric pulled back by the Lorentz/Möbius transformation that takes P/m to the rest observer. Equivalently, $F(n)/m$ is the standard boundary weight associated with

the interior point P/m of hyperbolic observer space. Thus (34) may be written simply as $f_{h,P} = -h dA_P$, where dA_P is the area form of (36).

Two consequences are useful. First, $P \mapsto \alpha P$ leaves m^2/F^2 unchanged, so the boundary Chern geometry depends only on the projective timelike direction and not on the radial normalization of the total charge. Second, as a timelike point approaches a projective null direction, the visual measure concentrates while its total area remains 4π ; the Chern flux therefore remains fixed even though its density becomes increasingly localized. This is complementary to the curvature-dilution picture on constant-staticity spheres developed in the spin paper.

The spinor metric construction does not identify the auxiliary Sen–Witten spinor with a physical helicity state. It only shows that both structures use the same projective spinor boundary: the positive-energy problem requires a bulk lift of a boundary spinor line, while physical helicity uses associated powers of that line.

6 Quantum staticity and consistency tests

We now restore c and $\hbar$. The aim is not to quantize the ADM boundary charge. Instead we record exact operator identities on specified positive-energy models and then isolate obstructions to importing the classical cone picture into an unconstrained Weyl representation.

6.1 A matched massive shell

Fix $Q > 0$ and let $\mathcal{H}_1 = L^2(\mathbb{R}^3, d^3p; \mathbb{C}^r)$, where r is an optional finite internal multiplicity. Let $\hat{p}_i$ be multiplication operators and define the positive-energy shell

$$\hat{E} = c\sqrt{Q^2\mathbf{1} + \hat{p}^2}, \quad \hat{D}_i = c\hat{p}_i\hat{E}^{-1}, \quad \hat{S} = cQ\hat{E}^{-1}, \quad \hat{q} = Q\hat{S} = cQ^2\hat{E}^{-1}. \quad (37)$$

These are strongly commuting spectral functions and satisfy $\sum_i \hat{D}_i^2 + \hat{S}^2 = \mathbf{1}$ and $c\hat{q} = \hat{E}\hat{S}^2$.

Proposition 6.1 (Matched-shell staticity bound). *For every Euclidean unit vector n , define $\hat{F}_n = \hat{E} - cn \cdot \hat{p}$. On the finite-energy quadratic-form domain,*

$$\hat{F}_n = \frac{c}{2}\hat{q} + \frac{\hat{E}}{2} \sum_i (\hat{D}_i - n_i\mathbf{1})^2 \geq \frac{c}{2}\hat{q}. \quad (38)$$

The coefficient 1/2 is optimal for a bound uniform in momentum and direction.

Proof. The commuting joint functions obey $\sum_i (D_i - n_i)^2 = 2 - 2n \cdot D - S^2$. Multiplying by $E/2$ and adding $cq/2 = ES^2/2$ gives (38). For momenta parallel to n with $|p|/Q \rightarrow \infty$, the scalar ratio $F_n/(cq)$ tends to $1/2$, so no larger uniform coefficient is possible. $\square$

The inequality applies to coherent superpositions in momentum space because it is an operator-form identity, not merely a classical convexity statement. It nevertheless assumes the positive massive shell from the start; it is not derived from the local soldering Weyl algebra.

6.2 Collective staticity

For independent positive-energy particles, let $E_T = \sum_\alpha E_\alpha$ and $P_T = \sum_\alpha p_\alpha$. On the nonvacuum sector define

$$D = \frac{cP_T}{E_T}, \quad Q_T^2 = (E_T/c)^2 - |P_T|^2, \quad q_T = \frac{cQ_T^2}{E_T}, \quad q_{\text{add}} = \sum_\alpha q_\alpha. \quad (39)$$

Strong commutativity permits the same definitions by joint functional calculus in the free quantum theory.

Proposition 6.2 (Collective-staticity variance). *The collective and additive staticities satisfy*

$$c(q_T - q_{\text{add}}) = \sum_{\alpha} E_{\alpha} |D_{\alpha} - D|^2 \geq 0. \quad (40)$$

Hence the total invariant scale contains relative-motion contributions and is generally not the sum of the one-particle staticities.

Proof. Use $cq_{\alpha} = E_{\alpha}(1 - |D_{\alpha}|^2)$ and $cq_T = E_T(1 - |D|^2)$ and expand the weighted variance around D . $\square$

The statement concerns additive free translations. It should not be used for a gravitating bound system by replacing the interacting total energy with a free sum.

6.3 A controlled attractive model

A simple interacting example shows how the staticity bound can survive negative binding energy when the internal mass operator remains positive. For two equal masses with $Q = mc$, consider on $L^2(\mathbb{R}^3, d^3r)$

$$\hat{H}_{\text{int}} = 2c\sqrt{\hat{p}^2 + Q^2} - \frac{GQ^2}{c^2 r}, \quad \alpha_G = \frac{Gm^2}{\hbar c}. \quad (41)$$

The sharp Kato–Herbst form inequality gives the standard semirelativistic stability bound [Herbst1977, HallLucha2006].

Proposition 6.3 (Attractive internal scale). *If $0 \leq \alpha_G < 4/\pi$, then the closed quadratic form associated with (41) satisfies*

$$\hat{H}_{\text{int}} \geq 2cQ\sqrt{1 - (\pi\alpha_G/4)^2} \mathbf{1} > 0. \quad (42)$$

If $\hat{Q}_{\text{int}} = \hat{H}_{\text{int}}/c$ strongly commutes with an external total momentum $\hat{P}$, then with $\hat{E}_T = c\sqrt{\hat{Q}_{\text{int}}^2 + \hat{P}^2}$ and $\hat{q}_T = c\hat{Q}_{\text{int}}^2\hat{E}_T^{-1}$ one has $\hat{E}_T - cn \cdot \hat{P} \geq (c/2)\hat{q}_T \geq 0$.

This is a restricted model, not a quantum Einstein system. Interacting internal-mass constructions in relativistic dynamics long predate the present MS formulation [BakamjianThomas1953].

6.4 Why a positive boundary energy cannot inherit full additive Weyl shifts

The local quantum-configuration analysis of MS soldering uses Weyl-type comparison relations. A positive physical boundary energy cannot simply be identified with an unconstrained generator carrying the same full additive shift symmetry.

Proposition 6.4 (Shift-covariance obstruction). *Let B be self-adjoint and suppose a unitary group satisfies*

$$U(s)BU(s)^{-1} = B + s\beta\mathbf{1}, \quad s \in \mathbb{R}, \quad \beta \neq 0. \quad (43)$$

Then the nonempty spectrum of B is invariant under every real translation and therefore equals $\mathbb{R}$. In particular, B cannot be bounded below.

Proof. Unitary equivalence preserves the spectrum, while (43) shifts it by $s\beta$. Thus $\sigma(B) = \sigma(B) + s\beta$ for every real s . $\square$

Any future quantum ADM completion must therefore alter the ambient comparison structure through constraints, a nonlinear positive observable, a restricted representation, an affine positive-variable quantization, or another compatible mechanism. Affine quantization provides one established example of changing the canonical kinematics when positivity is essential [Klauder1999].

6.5 Ordering and local-energy obstructions

Classical positivity of a polynomial symbol is not preserved by arbitrary quantization orderings. For $[\hat{x}, \hat{p}] = i\hbar \mathbf{1}$ and $\hat{B} = (\hat{x}\hat{p} + \hat{p}\hat{x})/2$, Weyl quantization gives

$$\text{Op}_W((xp)^2) = \hat{B}^2 - \frac{\hbar^2}{4} \mathbf{1}. \quad (44)$$

Since $\hat{B}$ has spectrum $\mathbb{R}$ in the standard representation, the right-hand side has negative spectrum despite $(xp)^2 \geq 0$. The classical frame identity therefore cannot be turned into a quantum positivity theorem by placing hats on its squares. An $A^\dagger A$ ordering would enforce form positivity but would still have to be shown to equal the correct boundary generator on the physical constraint space.

A second obstruction comes from quantum matter. Pointwise classical DEC cannot be promoted to a universal positivity statement for local quantum energy densities. Superpositions can have negative local expectation values even when the global Hamiltonian is vacuum-calibrated and nonnegative; quantum energy inequalities control suitable averages rather than impose a pointwise DEC [FewsterEveson1998]. Thus the classical term $T(N, \ell) \geq 0$ in (22) is not a ready-made quantum operator inequality.

6.6 A conditional spectral criterion

One general observation survives quantization at the representation-theoretic level. Suppose $(\hat{E}, \hat{P}_i)$ are strongly commuting self-adjoint operators on a physical Hilbert space and transform exactly as a four-vector under a unitary representation of the proper orthochronous Lorentz group, including their joint spectral domains. If $\hat{E}$ has a finite lower bound, Lorentz invariance of the joint spectrum excludes spacelike and past-directed spectral points: any such point can be boosted to arbitrarily negative energy. Hence the joint spectrum lies in the closed future cone and $\hat{E} - cn \cdot \hat{P} \geq 0$ by functional calculus.

This standard spectrum-condition argument is consistent with Wigner’s classification of positive-energy representations [Wigner1939]. It is only a criterion. For MS the unresolved tasks are to construct the physical constrained Hilbert space, define the actual asymptotic generators, prove their covariance and stability, and control boundary regularization and ordering. Restricted or formal quantum-positive-energy constructions already exist in canonical gravity and must be distinguished from any future MS result [Thiemann1998, Smolin2014].

7 Discussion

The normalized Lorentz–Hermitian cone provides a compact way to separate several statements that otherwise look similar. The rank-one boundary $\mathbb{CP}^1$ consists of projective null probes. A translation covector defines a Hermitian form on those probes, and positivity on every extreme ray is equivalent to membership of the total covector in the future cone. Massive observer positivity then follows from the convex decomposition of the closed MS ball. None of this uses gravitational dynamics.

Dynamics enters when one asks why the ADM charge of a physical initial data set satisfies those inequalities. The conformal identity (16) gives a direct proof on a useful restricted class. The frame current (22) removes the local sign obstructions without conformal flatness or maximality, while the small-data theorem shows that its gauge is not empty. The exact identification with the Sen–Witten equation then places the construction inside the established Witten/Nester positivity mechanism rather than beside it.

The selection problem is more informative than a simple relabeling. Universal positivity within a fixed current ansatz picks out the Sen–Witten trace coefficient, but the phase counterexample proves that projective null-current data do not determine the full spinor lift. The primitive MS

response therefore supplies the boundary ray but not, by itself, the phase-sensitive elliptic section required by the Witten argument. Any stronger derivation would need additional comparison geometry or a genuinely spinorial response principle.

The boundary Chern calculation sharpens the relation to the companion spin paper. A future-timelike charge induces a positive Hermitian metric on the basic spinor line, and the same null response $F(n)$ determines how the standard Chern curvature is redistributed over the projective sphere. The total Chern class is unchanged. This is a geometric bridge between the positive cone and helicity topology, not an identification of the Witten auxiliary spinor with a physical one-particle helicity state.

The quantum discussion points in the same direction. On a chosen positive shell the staticity identities are exact and survive simple composition and a controlled attractive internal-mass model. Yet full additive Weyl shifts are incompatible with semibounded energy, ordering can destroy classical square positivity, and local classical energy conditions do not become universal quantum operator inequalities. A quantum MS theory of asymptotic energy therefore cannot be obtained by transferring the ambient local Weyl algebra directly to the boundary. Constraint reduction, the physical inner product, asymptotic covariance, vacuum normalization and the boundary limit must be part of the construction.

Within the three-paper organization of the present program, the first paper constructs classical gravitational boundary charges from the MS-reconstructed geometry, the second develops the normalized cone and its spin/helicity boundary geometry, and the present paper addresses the positive-energy bridge between them. A general positive-mass theorem is not reproved here, and a quantum ADM operator is not constructed. The concrete results are instead the cone-duality formulation, the explicit classical response identities, the small-data frame construction, the restricted coefficient-selection formula, the phase-blind obstruction, the response-weighted Chern curvature, and the quantum staticity and no-go identities. Together they specify more sharply which additional structures are required for a genuinely independent MS dynamics.